

Sql script

```
/* -----
```

0. DATABASE

```
----- */
```

IF DB_ID(N'RaceDayDB') IS NOT NULL

BEGIN

ALTER DATABASE RaceDayDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;

DROP DATABASE RaceDayDB;

END

GO

CREATE DATABASE RaceDayDB;

GO

USE RaceDayDB;

GO

```
/* -----
```

1. TABLE: Users

Participants, organisers, and admins all live here; Role distinguishes them.

```
----- */
```

CREATE TABLE dbo.Users

(

UserID INT IDENTITY(1,1) NOT NULL,

FullName NVARCHAR(100) NOT NULL,

```

Email      NVARCHAR(150)      NOT NULL,
Phone      NVARCHAR(20)       NULL,
PasswordHash NVARCHAR(255)    NOT NULL,
Role       NVARCHAR(20)       NOT NULL
          CONSTRAINT DF_Users_Role DEFAULT ('Participant'),
DateCreated DATETIME2         NOT NULL
          CONSTRAINT DF_Users_DateCreated DEFAULT (SYSUTCDATETIME()),

CONSTRAINT PK_Users PRIMARY KEY CLUSTERED (UserID),
CONSTRAINT UQ_Users_Email UNIQUE (Email),
CONSTRAINT CK_Users_Role CHECK (Role IN (N'Participant', N'Organiser', N'Admin'))
);
GO

```

```

/* -----

```

2. TABLE: Organisations

A club / race committee that hosts events. Each is managed by one accountable User (an Organiser).

```

----- */

```

```

CREATE TABLE dbo.Organisations
(
    OrganisationID INT IDENTITY(1,1) NOT NULL,
    ManagedByUserID INT NOT NULL,
    Name NVARCHAR(150) NOT NULL,
    ContactEmail NVARCHAR(150) NOT NULL,
    ContactPhone NVARCHAR(20) NULL,

```

DateCreated DATETIME2 NOT NULL

CONSTRAINT DF_Organisations_DateCreated DEFAULT (SYSUTCDATETIME()),

CONSTRAINT PK_Organisations PRIMARY KEY CLUSTERED (OrganisationID),

CONSTRAINT UQ_Organisations_ContactEmail UNIQUE (ContactEmail),

CONSTRAINT FK_Organisations_Users FOREIGN KEY (ManagedByUserID)

REFERENCES dbo.Users (UserID)

ON DELETE NO ACTION

ON UPDATE NO ACTION

);

GO

/* -----

3. TABLE: Venues

Physical locations events are hosted at (also used for
live weather lookups by coordinates).

----- */

CREATE TABLE dbo.Venues

(

VenueID INT IDENTITY(1,1) NOT NULL,

Name NVARCHAR(150) NOT NULL,

Address NVARCHAR(255) NULL,

City NVARCHAR(100) NOT NULL,

Province NVARCHAR(100) NOT NULL,

Latitude DECIMAL(9,6) NULL,

Longitude DECIMAL(9,6) NULL,

```
CONSTRAINT PK_Venues PRIMARY KEY CLUSTERED (VenueID)
```

```
);
```

```
GO
```

```
/* -----
```

```
4. TABLE: Events
```

```
----- */
```

```
CREATE TABLE dbo.Events
```

```
(
```

```
EventID INT IDENTITY(1,1) NOT NULL,
```

```
OrganisationID INT NOT NULL,
```

```
VenueID INT NOT NULL,
```

```
Name NVARCHAR(150) NOT NULL,
```

```
EventDate DATE NOT NULL,
```

```
Discipline NVARCHAR(20) NOT NULL,
```

```
Description NVARCHAR(MAX) NULL,
```

```
DateCreated DATETIME2 NOT NULL
```

```
CONSTRAINT DF_Events_DateCreated DEFAULT (SYSUTCDATETIME()),
```

```
CONSTRAINT PK_Events PRIMARY KEY CLUSTERED (EventID),
```

```
CONSTRAINT FK_Events_Organisations FOREIGN KEY (OrganisationID)
```

```
REFERENCES dbo.Organisations (OrganisationID)
```

```
ON DELETE NO ACTION
```

```
ON UPDATE NO ACTION,
```

```
CONSTRAINT FK_Events_Venues FOREIGN KEY (VenueID)
```

```

REFERENCES dbo.Venues (VenueID)

ON DELETE NO ACTION

ON UPDATE NO ACTION,

CONSTRAINT CK_Events_Discipline CHECK (Discipline IN (N'Running', N'Cycling',
N'Walking'))

);

GO

```

```

/* -----

```

5. TABLE: Routes

Each Event has one or more Routes; Categories later pick which Route they follow.

```

----- */

```

```

CREATE TABLE dbo.Routes

```

```

(

```

```

    RouteID    INT          IDENTITY(1,1) NOT NULL,

```

```

    EventID    INT          NOT NULL,

```

```

    Name       NVARCHAR(150) NOT NULL,

```

```

    DistanceKM DECIMAL(6,2)  NOT NULL,

```

```

    ElevationGain DECIMAL(7,2) NOT NULL

```

```

        CONSTRAINT DF_Routes_ElevationGain DEFAULT (0),

```

```

    SurfaceType NVARCHAR(50)  NULL,

```

```

    GPXFileURL  NVARCHAR(500) NULL,

```

```

    CONSTRAINT PK_Routes PRIMARY KEY CLUSTERED (RouteID),

```

```

    CONSTRAINT FK_Routes_Events FOREIGN KEY (EventID)

```

```

REFERENCES dbo.Events (EventID)

ON DELETE NO ACTION

ON UPDATE NO ACTION,

CONSTRAINT CK_Routes_DistanceKM CHECK (DistanceKM > 0)

);

GO

```

```

/* -----

```

6. TABLE: Categories

e.g. "10km Open", "21km Men 30-39" — belongs to one Event
and follows one Route.

```

----- */

```

```

CREATE TABLE dbo.Categories

```

```

(

```

```

    CategoryID    INT          IDENTITY(1,1) NOT NULL,

```

```

    EventID       INT          NOT NULL,

```

```

    RouteID       INT          NOT NULL,

```

```

    Name          NVARCHAR(100) NOT NULL,

```

```

    DistanceKM     DECIMAL(6,2) NOT NULL,

```

```

    MinAge         INT          NOT NULL

```

```

        CONSTRAINT DF_Categories_MinAge DEFAULT (0),

```

```

    MaxAge         INT          NOT NULL

```

```

        CONSTRAINT DF_Categories_MaxAge DEFAULT (99),

```

```

    Gender         NVARCHAR(10) NOT NULL

```

```

        CONSTRAINT DF_Categories_Gender DEFAULT (N'Open'),

```

```

    EntryFee       DECIMAL(8,2) NOT NULL

```

```

        CONSTRAINT DF_Categories_EntryFee DEFAULT (0),

CONSTRAINT PK_Categories PRIMARY KEY CLUSTERED (CategoryID),
CONSTRAINT FK_Categories_Events FOREIGN KEY (EventID)
    REFERENCES dbo.Events (EventID)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION,
CONSTRAINT FK_Categories_Routes FOREIGN KEY (RouteID)
    REFERENCES dbo.Routes (RouteID)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION,
CONSTRAINT CK_Categories_Gender CHECK (Gender IN (N'Male', N'Female', N'Open')),
CONSTRAINT CK_Categories_EntryFee CHECK (EntryFee >= 0),
CONSTRAINT CK_Categories_AgeRange CHECK (MaxAge >= MinAge)
);
GO

```

```

/* -----

```

7. TABLE: Registrations

A participant (User) entering a Category. One User may not enter the same Category twice.

```

----- */

```

```

CREATE TABLE dbo.Registrations
(

```

```

    RegistrationID    INT          IDENTITY(1,1) NOT NULL,
    UserID            INT          NOT NULL,

```

```

CategoryID      INT              NOT NULL,
RegistrationDate DATETIME2        NOT NULL

                CONSTRAINT DF_Registrations_RegistrationDate DEFAULT
(SYSUTCDATETIME()),
BibNumber       NVARCHAR(10)      NULL,
PaymentStatus   NVARCHAR(20)      NOT NULL

                CONSTRAINT DF_Registrations_PaymentStatus DEFAULT (N'Pending'),
EmergencyContactName NVARCHAR(100) NULL,
EmergencyContactPhone NVARCHAR(20) NULL,

CONSTRAINT PK_Registrations PRIMARY KEY CLUSTERED (RegistrationID),
CONSTRAINT FK_Registrations_Users FOREIGN KEY (UserID)

REFERENCES dbo.Users (UserID)

ON DELETE NO ACTION

ON UPDATE NO ACTION,
CONSTRAINT FK_Registrations_Categories FOREIGN KEY (CategoryID)

REFERENCES dbo.Categories (CategoryID)

ON DELETE NO ACTION

ON UPDATE NO ACTION,
CONSTRAINT UQ_Registrations_User_Category UNIQUE (UserID, CategoryID),
CONSTRAINT CK_Registrations_PaymentStatus CHECK (PaymentStatus IN (N'Pending',
N'Confirmed', N'Refunded', N'Cancelled'))
);
GO

```

/* -----

8. TABLE: Results

At most one Result per Registration (0..1) — populated once the participant finishes (or is marked DNF/DNS/DQ).

----- */

```
CREATE TABLE dbo.Results
```

```
(
```

```
    ResultID      INT      IDENTITY(1,1) NOT NULL,
```

```
    RegistrationID INT      NOT NULL,
```

```
    FinishTime    TIME(0)    NULL,
```

```
    ChipTime      TIME(0)    NULL,
```

```
    OverallPosition INT      NULL,
```

```
    CategoryPosition INT      NULL,
```

```
    Status        NVARCHAR(20) NOT NULL
```

```
        CONSTRAINT DF_Results_Status DEFAULT (N'Finished'),
```

```
    CONSTRAINT PK_Results PRIMARY KEY CLUSTERED (ResultID),
```

```
    CONSTRAINT FK_Results_Registrations FOREIGN KEY (RegistrationID)
```

```
        REFERENCES dbo.Registrations (RegistrationID)
```

```
        ON DELETE CASCADE
```

```
        ON UPDATE NO ACTION,
```

```
    CONSTRAINT UQ_Results_RegistrationID UNIQUE (RegistrationID),
```

```
    CONSTRAINT CK_Results_Status CHECK (Status IN (N'Finished', N'DNF', N'DNS', N'DQ')),
```

```
    CONSTRAINT CK_Results_Positions CHECK (OverallPosition IS NULL OR OverallPosition > 0)
```

```
);
```

```
GO
```

/* =====

SEED DATA

===== */

/* -----

Users: 2 Organisers, 2 Participants

----- */

INSERT INTO dbo.Users (FullName, Email, Phone, PasswordHash, Role)

VALUES

(N'Thandiwe Mokoena', N'thandiwe.mokoena@comradesrun.co.za', N'0827761234',
N'\$2a\$12\$hash_placeholder_org1', N'Organiser'),

(N'Pieter van der Merwe', N'pieter@capecycletour.co.za', N'0839912345',
N'\$2a\$12\$hash_placeholder_org2', N'Organiser'),

(N'Lindiwe Dlamini', N'lindiwe.dlamini@gmail.com', N'0716654321',
N'\$2a\$12\$hash_placeholder_p1', N'Participant'),

(N'Johan Botha', N'johan.botha@outlook.com', N'0721239876',
N'\$2a\$12\$hash_placeholder_p2', N'Participant');

GO

/* -----

Organisations: 2, each managed by one of the organisers above

----- */

INSERT INTO dbo.Organisations (ManagedByUserID, Name, ContactEmail, ContactPhone)

VALUES

((SELECT UserID FROM dbo.Users WHERE Email =
N'thandiwe.mokoena@comradesrun.co.za'),

N'Comrades Marathon Association', N'info@comradesrun.co.za', N'0331234567'),

```
((SELECT UserID FROM dbo.Users WHERE Email = N'pieter@capecycletour.co.za'),  
N'Cape Town Cycle Tour Trust', N'office@capecycletour.co.za', N'0214567890');  
GO
```

```
/* -----
```

Venues: one per event

```
----- */
```

```
INSERT INTO dbo.Venues (Name, Address, City, Province, Latitude, Longitude)
```

```
VALUES
```

```
(N'Pietermaritzburg City Hall', N'333 Church St', N'Pietermaritzburg', N'KwaZulu-Natal', -  
29.600950, 30.379630),
```

```
(N'Grand Parade', N'Darling St', N'Cape Town', N'Western Cape', -33.925280,  
18.424140),
```

```
(N'FNB Stadium', N'Nasrec Rd', N'Johannesburg', N'Gauteng', -26.234800,  
27.982400);
```

```
GO
```

```
/* -----
```

Events: 3

```
----- */
```

```
INSERT INTO dbo.Events (OrganisationID, VenueID, Name, EventDate, Discipline,  
Description)
```

```
VALUES
```

```
((SELECT OrganisationID FROM dbo.Organisations WHERE Name = N'Comrades  
Marathon Association'),
```

```
(SELECT VenueID FROM dbo.Venues WHERE Name = N'Pietermaritzburg City Hall'),
```

```
N'Comrades Marathon', '2027-06-13', N'Running',
```

```

        N'Iconic ultramarathon between Pietermaritzburg and Durban.'),
        ((SELECT OrganisationID FROM dbo.Organisations WHERE Name = N'Cape Town Cycle
Tour Trust'),
        (SELECT VenueID FROM dbo.Venues WHERE Name = N'Grand Parade'),
        N'Cape Town Cycle Tour', '2027-03-08', N'Cycling',
        N'World-famous timed cycle race around the Cape Peninsula.'),
        ((SELECT OrganisationID FROM dbo.Organisations WHERE Name = N'Comrades
Marathon Association'),
        (SELECT VenueID FROM dbo.Venues WHERE Name = N'FNB Stadium'),
        N'Soweto Marathon', '2027-11-07', N'Running',
        N'Community road race through the streets of Soweto.');
```

GO

```

/* -----
```

Routes: at least one per event

```

----- */
```

```

INSERT INTO dbo.Routes (EventID, Name, DistanceKM, ElevationGain, SurfaceType,
GPXFileURL)
```

```

VALUES
```

```

        ((SELECT EventID FROM dbo.Events WHERE Name = N'Comrades Marathon'),
        N'Up Run - PMB to Durban', 87.00, 1200.00, N'Tar road',
        N'https://raceday.co.za/gpx/comrades-up.gpx'),
        ((SELECT EventID FROM dbo.Events WHERE Name = N'Cape Town Cycle Tour'),
        N'Full Peninsula Loop', 109.00, 1000.00, N'Tar road', N'https://raceday.co.za/gpx/ctct-
full.gpx'),
        ((SELECT EventID FROM dbo.Events WHERE Name = N'Cape Town Cycle Tour'),
        N'Mini Peninsula Loop', 43.00, 350.00, N'Tar road', N'https://raceday.co.za/gpx/ctct-
mini.gpx'),
```

```
((SELECT EventID FROM dbo.Events WHERE Name = N'Soweto Marathon'),
    N'Soweto 42.2km Route', 42.20, 300.00, N'Tar road',
N'https://raceday.co.za/gpx/soweto-42.gpx'),
    ((SELECT EventID FROM dbo.Events WHERE Name = N'Soweto Marathon'),
    N'Soweto 10km Route', 10.00, 80.00, N'Tar road', N'https://raceday.co.za/gpx/soweto-
10.gpx');
GO
```

```
/* -----
```

Categories: at least one per event, following the routes above

```
----- */
```

```
INSERT INTO dbo.Categories (EventID, RouteID, Name, DistanceKM, MinAge, MaxAge,
Gender, EntryFee)
```

```
VALUES
```

```
((SELECT EventID FROM dbo.Events WHERE Name = N'Comrades Marathon'),
    (SELECT RouteID FROM dbo.Routes WHERE Name = N'Up Run - PMB to Durban'),
    N'Comrades Open', 87.00, 20, 99, N'Open', 950.00),
    ((SELECT EventID FROM dbo.Events WHERE Name = N'Comrades Marathon'),
    (SELECT RouteID FROM dbo.Routes WHERE Name = N'Up Run - PMB to Durban'),
    N'Comrades Veteran 50+', 87.00, 50, 99, N'Open', 950.00),
    ((SELECT EventID FROM dbo.Events WHERE Name = N'Cape Town Cycle Tour'),
    (SELECT RouteID FROM dbo.Routes WHERE Name = N'Full Peninsula Loop'),
    N'Full Loop Open', 109.00, 18, 99, N'Open', 650.00),
    ((SELECT EventID FROM dbo.Events WHERE Name = N'Cape Town Cycle Tour'),
    (SELECT RouteID FROM dbo.Routes WHERE Name = N'Mini Peninsula Loop'),
    N'Mini Loop Open', 43.00, 14, 99, N'Open', 350.00),
    ((SELECT EventID FROM dbo.Events WHERE Name = N'Soweto Marathon'),
```

```

        (SELECT RouteID FROM dbo.Routes WHERE Name = N'Soweto 42.2km Route'),
        N'Marathon Open', 42.20, 20, 99, N'Open', 400.00),
    ((SELECT EventID FROM dbo.Events WHERE Name = N'Soweto Marathon'),
        (SELECT RouteID FROM dbo.Routes WHERE Name = N'Soweto 10km Route'),
        N'10km Fun Run', 10.00, 10, 99, N'Open', 150.00);

GO

```

```

/* -----

```

Registrations: sample enrolments from the 2 participants

```

----- */

```

```

INSERT INTO dbo.Registrations (UserID, CategoryID, BibNumber, PaymentStatus,
EmergencyContactName, EmergencyContactPhone)

```

```

VALUES

```

```

    ((SELECT UserID FROM dbo.Users WHERE Email = N'lindiwe.dlamini@gmail.com'),
        (SELECT CategoryID FROM dbo.Categories WHERE Name = N'Comrades Open'),
        N'A1023', N'Confirmed', N'Nomsa Dlamini', N'0827771122'),
    ((SELECT UserID FROM dbo.Users WHERE Email = N'lindiwe.dlamini@gmail.com'),
        (SELECT CategoryID FROM dbo.Categories WHERE Name = N'10km Fun Run'),
        N'S0456', N'Confirmed', N'Nomsa Dlamini', N'0827771122'),
    ((SELECT UserID FROM dbo.Users WHERE Email = N'johan.botha@outlook.com'),
        (SELECT CategoryID FROM dbo.Categories WHERE Name = N'Full Loop Open'),
        N'C2091', N'Confirmed', N'Elsie Botha', N'0839981122'),
    ((SELECT UserID FROM dbo.Users WHERE Email = N'johan.botha@outlook.com'),
        (SELECT CategoryID FROM dbo.Categories WHERE Name = N'Marathon Open'),
        N'S0789', N'Pending', N'Elsie Botha', N'0839981122');

```

```

GO

```

```
/* -----
```

Results: sample finish data for the confirmed entries

```
----- */
```

```
INSERT INTO dbo.Results (RegistrationID, FinishTime, ChipTime, OverallPosition,  
CategoryPosition, Status)
```

```
VALUES
```

```
((SELECT RegistrationID FROM dbo.Registrations WHERE BibNumber = N'A1023'),  
 '10:42:15', '10:41:02', 1845, 612, N'Finished'),
```

```
((SELECT RegistrationID FROM dbo.Registrations WHERE BibNumber = N'S0456'),  
 '00:52:30', '00:52:10', 88, 12, N'Finished'),
```

```
((SELECT RegistrationID FROM dbo.Registrations WHERE BibNumber = N'C2091'),  
 '03:15:40', '03:15:05', 420, 95, N'Finished');
```